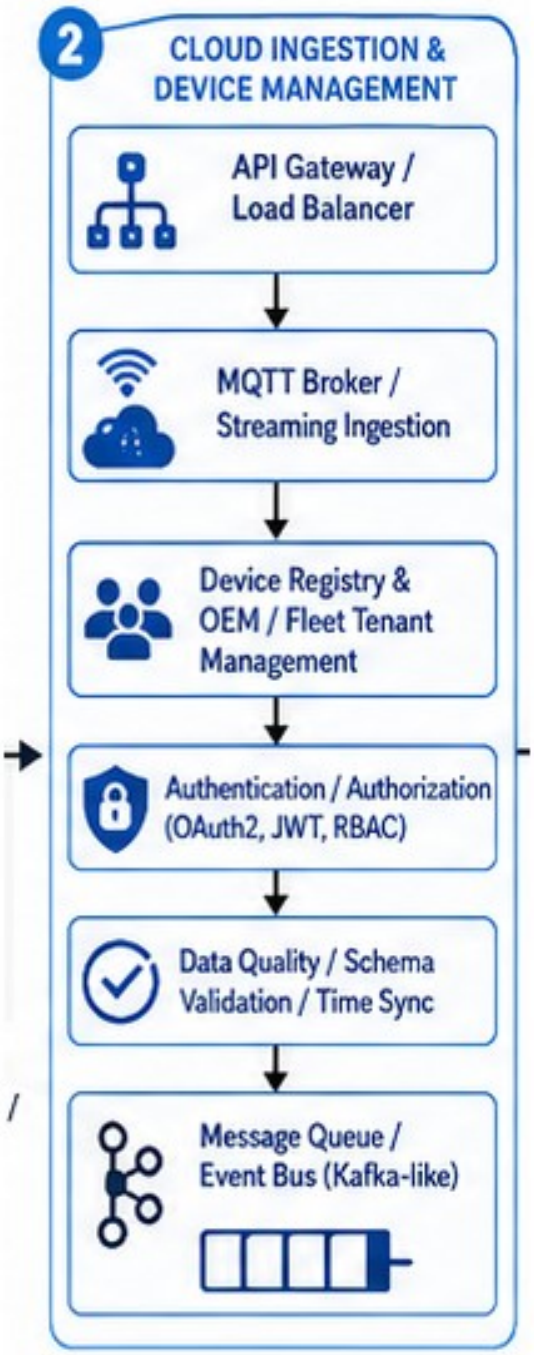
Step 2 - Cloud Ingestion & Device Management Layer

Development, Testing, Validation and Acceptance Guide


2
CLOUD INGESTION &
DEVICE MANAGEMENT
API Gateway /
Load Balancer
MQTT Broker /
Streaming Ingestion
Device Registry &
OEM / Fleet Tenant
Management
Authentication / Authorization
(OAuth2, JWT, RBAC)
Data Quality / Schema
Validation / Time Sync
Message Queue /
Event Bus (Kafka-like)

Development Workflow | Testing & Validation | Acceptance Criteria | Timeline | Reference Material

Revision 1 | August 2026

Document Purpose and Structure

This guide converts the platform architecture block into an implementable engineering work package. It defines the functional scope, interfaces, development activities, verification strategy, acceptance criteria, evidence required for sign-off, delivery timeline, and reference resources.

All numerical acceptance values in this guide are baseline project targets for the first controlled release. They should be recalibrated after representative load testing, vehicle/telematics characterization, and customer/OEM requirements are frozen.

Document structure

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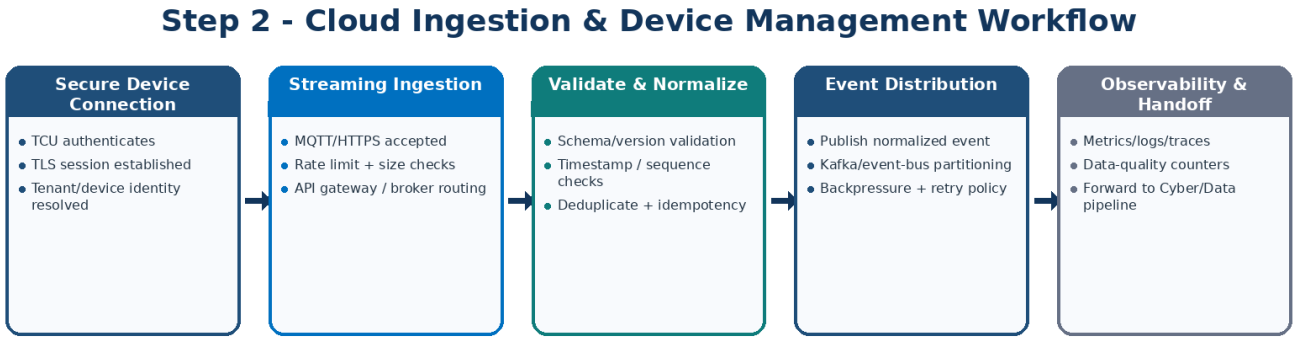
1. Role and Scope

Step 2 is the controlled entry point for vehicle and battery telemetry into the cloud platform. Its purpose is to establish an authenticated device session, accept telemetry at scale, normalize it into a stable internal event contract, enforce tenant/device context, and publish trusted ingestion events to downstream cybersecurity and battery-data services.

Design rule: Step 2 must not make battery-safety decisions. It guarantees transport, identity, ordering/quality metadata, and reliable handoff. Cyber/physics trust classification remains owned by Step 3.

In scope	Out of scope
API gateway / load balancing; MQTT/HTTPS ingestion; device registry; tenant mapping; OAuth2/JWT/RBAC integration; schema/time validation; event bus; retry/backpressure; observability	Cyber IDS model, thermal AI, blockchain notarization, RAG reasoning, final web-dashboard rendering
Telemetry admission and normalized event envelope	Vehicle control commands or safety-critical closed-loop actuation from cloud
Device lifecycle onboarding, disable/revoke, metadata and fleet assignment	Native mobile application development

2. End-to-End Workflow



1. TCU/Gateway presents its device identity and opens a TLS-protected MQTT or HTTPS connection.
2. API gateway or broker validates connection policy, tenant/device registration, rate limits and allowed topics/endpoints.
3. The message is decoded against the currently approved schema version and receives ingestion metadata such as cloud receive time, sequence status and quality flags.
4. Duplicate/idempotent processing, timestamp checks and payload-size/range sanity checks are applied before downstream publication.
5. Normalized events are published to the event bus using deterministic topic/partition keys such as tenant, vehicle and battery ID.
6. Metrics, audit logs and distributed traces are emitted for operational monitoring. Downstream Step 3 receives the normalized stream for trust classification.

3. Input Message and Data Contract

Field	Purpose	Validation
tenant_id	OEM/fleet ownership and isolation context	Must match authenticated device assignment
device_id / vehicle_id / battery_id	Stable identity and routing keys	Registered, active and relationship-valid
event_time_utc	Vehicle/TCU measurement timestamp	UTC, monotonic policy, skew window
sequence_no	Replay, gap and ordering detection	Monotonic per configured stream
schema_version	Payload compatibility	Must be supported; unknown version quarantined/rejected
signals	CAN-decoded telemetry payload	Required fields, units, datatype and size validated
edge_quality	TCU-side data quality and freshness metadata	Enumerated / bounded
ingest_time_utc	Cloud receive timestamp	Added by ingestion service
correlation_id	Traceability across services	Generated/preserved end-to-end

Recommended event practice: preserve the original event time and add an independent cloud ingestion time.

Never overwrite source timestamps because latency, replay and data-quality analysis require both.

4. API Gateway and Load Balancing

- **Protocol termination:** Expose approved HTTPS endpoints and/or route MQTT broker connections through controlled ingress.
- **Rate enforcement:** Apply per-device, per-tenant and global request/message ceilings to protect shared services.
- **Payload controls:** Enforce content type, maximum payload size, compression policy and supported schema versions.
- **Routing:** Route onboarding, telemetry, heartbeat and administrative traffic to separate backend services.
- **Protection:** Reject unauthenticated traffic before it reaches application services; integrate WAF/DDoS controls at the perimeter.
- **Health management:** Remove unhealthy instances from rotation and expose readiness/liveness checks.

5. MQTT / HTTPS Streaming Ingestion

Capability	MQTT path	HTTPS path
Typical use	Continuous telemetry / publish-subscribe	Batch upload, commands not used for safety, administrative APIs
Delivery semantics	QoS policy selected by topic; retain disabled for telemetry unless justified	HTTP status + client retry with idempotency key
Session identity	mTLS/client credential + topic ACL	OAuth2/mTLS/JWT as applicable
Backpressure	Broker quotas, inflight limits, queue depth	429/503 response with Retry-After and client retry policy
Observability	Broker connection/message metrics	Request/latency/error metrics

Do not assume MQTT QoS alone guarantees exactly-once business processing. Downstream idempotency and duplicate detection are still required.

6. Device Registry and Multi-Tenant Management

- Store device certificate/credential references, OEM/fleet tenant, vehicle, battery, firmware and allowed topic/API policy.
- Support lifecycle states: PROVISIONED -> ACTIVE -> SUSPENDED -> REVOKED -> RETIRED.
- Maintain device-to-vehicle and battery assignment history; do not silently overwrite historical ownership relationships.
- Every event must carry tenant context derived from authenticated identity, not from an untrusted payload field alone.
- Administrative changes require audit logging and role separation.

7. Authentication and Authorization

Actor	Authentication	Authorization
TCU / gateway	mTLS certificate or device credential	Topic/API ACL; assigned tenant + vehicle/battery only
OEM/Fleet admin	OIDC/OAuth2 interactive login	Tenant admin functions only
Backend service	Workload identity / short-lived token	Least-privilege service scopes
Support / operations	SSO + MFA	Controlled support role; audited access

8. Data Quality, Schema Validation and Time Synchronization

- **Schema:** Use versioned JSON/Protobuf/Avro schemas with explicit units and optional/required fields.
- **Timestamp:** Check stale/future timestamps, clock skew and impossible sequence regressions.
- **Completeness:** Track missing signals separately from physically invalid values.
- **Freshness:** Add age/quality flags so downstream models can decide whether an input is usable.
- **Units:** Normalize to canonical engineering units before Step 4 feature engineering.
- **Unknown fields:** Define forward/backward compatibility policy; avoid silently changing semantics.

9. Message Queue / Event Bus and Backpressure

- Partition primarily by tenant + vehicle/battery key to preserve ordered processing where required.
- Separate raw-ingest, validated-normalized, dead-letter and operational topics.
- Define retention based on recovery/replay needs rather than using the event bus as long-term storage.
- Use consumer groups for scalable downstream processing.
- Monitor consumer lag and apply load shedding or controlled degradation before queues exhaust resources.
- Dead-letter messages must retain original payload hash, rejection reason, schema version and correlation ID.

10. Reliability, Idempotency and Failure Handling

Failure / condition	Required behavior
Duplicate message	Detect by device/stream sequence and/or idempotency key; process business event once
Out-of-order event	Preserve source timestamp/sequence; mark ordering quality; downstream policy decides use
Broker/API unavailable	Client bounded retry with exponential backoff and local TCU buffering
Event bus unavailable	Ingress buffers within capacity; reject gracefully rather than unbounded memory growth
Unknown schema	Route to controlled rejection/dead-letter path with alert
Device revoked	New sessions denied; existing sessions terminated according to policy
Clock drift	Flag event quality; raise device health issue after calibrated threshold

11. Observability and Operations

Signal	Primary KPI / evidence
Traffic	messages/s, bytes/s, connected devices, publish success/failure
Latency	edge send -> ingest, ingest -> event bus, P50/P95/P99
Quality	schema reject rate, missing fields, timestamp skew, sequence gaps, duplicates
Reliability	retry rate, dead-letter count, consumer lag, reconnects
Security handoff	events forwarded to Step 3, quarantined events, identity failures
Resource	CPU, memory, network, broker disk/queue depth, autoscaling state

12. Development Lifecycle and Completion Timeline

Activity	Deliverable	Target timing	Exit condition
Scope, telemetry volume and tenant model	Context diagram, signal/event inventory, NFR baseline	Week 1	Architecture boundary and scale assumptions approved
Data contract and schema governance	Canonical envelope + versioning policy	W1-W2	Sample payloads and compatibility rules frozen
Device registry and identity prototype	Registry service + onboarding flow	W2-W3	Device lifecycle and tenant assignment demonstrated
MQTT/HTTPS ingress + gateway	Broker/API gateway baseline	W2-W4	Authenticated message enters controlled pipeline
Normalization + quality checks	Validation service + dead-letter flow	W3-W5	Invalid/schema/time cases handled deterministically
Event bus integration	Topics, partitions, consumer contracts	W4-W6	Stable handoff to Step 3/4 demonstrated
Reliability features	Retry, idempotency, backpressure	W5-W7	Failure injection shows controlled recovery
Observability	Metrics, logs, traces, dashboards	W6-W8	Root cause can be traced across ingest path
Automated V&V	Functional/API/integration/load test suite	W7-W10	Regression suite green
Scale/resilience test	Peak load + failure/chaos report	W10-W11	Capacity and recovery targets met
Pilot integration	Representative TCU/fleet pilot	W11-W12	Pilot evidence accepted
Release gate	Runbook, SBOM, config baseline, sign-offs	End W12	G7 approval

13. Verification and Validation Strategy

V&V layer	Method	Evidence
Unit	Parser, schema, registry, auth, retry/idempotency unit tests	Automated unit report + coverage
Contract	Producer/consumer schema and API contract tests	Contract compatibility report
Integration	MQTT/API -> normalization -> event bus -> Step 3 stub	End-to-end integration report
Performance	Sustained + burst telemetry, connection scale, consumer lag	Load report with latency/throughput/resource curves
Resilience	Broker restart, service crash, network loss, slow consumer	Recovery and data-loss evidence
Security	Unauthorized device, cross-tenant access,	Security test report

V&V layer	Method	Evidence
	token/cert expiry	
Data quality	Malformed schema, timestamp drift, missing/duplicate/out-of-order data	Quality handling matrix
Pilot	Representative TCU + actual network variability	Pilot acceptance report

14. Test Catalogue

Test ID	Scenario	Expected result
ING-01	Valid MQTT telemetry	Accepted, normalized and published once
ING-02	Unknown/revoked device	Connection/message rejected and audited
ING-03	Wrong tenant/topic	Authorization denied; no cross-tenant event
ING-04	Unsupported schema version	Rejected or dead-lettered with reason
ING-05	Duplicate sequence/idempotency key	Single business event downstream
ING-06	Out-of-order / gap	Preserved with quality/order metadata
ING-07	Clock skew / future timestamp	Quality flag + policy action
ING-08	Burst > nominal	Queue/backpressure protects services; no uncontrolled memory growth
ING-09	Broker restart	Clients recover; admitted buffered events processed per policy
ING-10	Slow downstream consumer	Lag visible; system remains bounded
ING-11	Expired cert/token	Authentication fails and is logged
ING-12	Network interruption	TCU buffers/retries; replay handled idempotently
ING-13	Malformed payload / oversized message	Rejected before downstream pipeline
ING-14	Multi-tenant load	No event/data leakage between tenants
ING-15	Observability correlation	Single correlation ID trace visible across services

15. Acceptance Criteria - Project Targets

Metric	Baseline project target	Acceptance method
Authorized telemetry admission	100% of valid registered-device test messages accepted	Automated positive-path test suite
Unauthorized / revoked device rejection	100% rejected	Negative authentication tests
Cross-tenant access	0 successful cross-tenant operations in test suite	Tenant-isolation security tests
Schema-invalid message handling	100% deterministically rejected/dead-lettered	Schema fault injection
Duplicate business processing	0 duplicate downstream events for replayed idempotency set	Replay / duplicate test
Admitted-message loss	0 unexplained loss during nominal and designed recovery tests	Sequence reconciliation

Metric	Baseline project target	Acceptance method
Ingest-to-event-bus latency	P95 <= 500 ms at nominal planned load	Load test
Peak headroom	Demonstrate >= 2x forecast nominal telemetry rate for 30 min	Performance test
Consumer lag recovery	Returns below operating threshold after controlled burst	Burst/recovery test
Observability completeness	>= 99% sampled end-to-end transactions correlate by trace/correlation ID	Trace audit
Critical vulnerabilities	0 open Critical/High findings at release	SAST/SCA/DAST/security closure
Recovery	Documented recovery completed within agreed pilot RTO/RPO	Failure/recovery exercise

16. Acceptance Gates and Sign-off

Gate	Review question	Required evidence
G0 - Scope Freeze	Are device populations, telemetry volume, protocols, tenants and boundaries defined?	Architecture, load assumptions, event inventory
G1 - Contract Freeze	Are event envelope, schemas, topic/API naming and compatibility rules frozen?	Schema repository, contract tests, sample payloads
G2 - Design Freeze	Are gateway, broker, registry, auth, event bus and failure flows specified?	Design + interface specifications
G3 - Component Ready	Do services pass unit/contract/security checks?	Unit, coverage, SAST/SCA, service readiness
G4 - Integrated Validation	Does authenticated telemetry flow end-to-end under normal and negative cases?	Integration and data-quality reports
G5 - Scale/Resilience	Do load, burst, broker/service failure and recovery targets pass?	Performance + resilience reports
G6 - Pilot Acceptance	Does representative TCU/fleet operation meet latency, quality and operational needs?	Pilot report + operations sign-off
G7 - Production Release	Are runbooks, monitoring, capacity, rollback, SBOM and sign-offs complete?	Release pack and approvals

17. Risks and Mitigations

Risk	Impact	Mitigation
Underestimated fleet volume / burstiness	Queue growth, latency, message rejection	Capacity model, 2x+ headroom test, partition/autoscale plan
Unstable telematics connectivity	Replay bursts and data gaps	Edge buffering, bounded retry, idempotency, timestamp/sequence metadata
Schema drift	Downstream parser/model failure	Registry, compatibility CI, versioned contracts
Tenant misconfiguration	Data isolation breach	Identity-derived tenant context, negative isolation tests
Clock quality	Misleading temporal analytics	NTP/GNSS strategy, skew flags, source + ingest timestamps
Slow consumer	Event-bus lag and resource exhaustion	Consumer lag alarms, bounded retention/backpressure
Certificate lifecycle failure	Fleet disconnect	Staged rotation, renewal monitoring, emergency recovery process

18. Engineering Deliverables

- Step-2 architecture and trust/data-flow diagrams
- Telemetry/event data dictionary and schema repository
- MQTT topic and HTTPS API specification
- Device registry and lifecycle specification
- Tenant/RBAC authorization matrix
- Retry, buffering, idempotency and dead-letter policy
- Event-bus topic/partition/retention specification
- Observability dashboard and alert definitions
- Automated unit/contract/integration/load/security test suite
- Performance/capacity and resilience report
- Pilot report
- Production runbook, SBOM, release manifest and gate sign-offs

19. Reference Material and Open-Source Validation Resources

[1] OASIS MQTT Version 5.0 Official specification

[2] Eclipse Mosquitto Open-source MQTT broker documentation

[3] Eclipse Paho Open-source MQTT client libraries

[4] Apache Kafka Event-streaming platform documentation

[5] Keycloak Open-source identity and access management

[6] OpenTelemetry Metrics, traces and logs documentation

[7] Grafana k6 Open-source load/performance testing

[8] IoT data dissemination scheme for reducing delay in multi-broker environments Internet of Things, 2024, 25:101025

[9] Improvement of MQTT semantic to minimize data flow in IoT platforms based on distributed brokers Transactions on Emerging Telecommunications Technologies, 2024

[10] IIoT Protocols for Edge/Fog and Cloud Computing in Industrial AI International Journal of Cloud Applications and Computing, 2024

[11] Dual Digital Twin: Cloud-edge collaboration with Lyapunov-based incremental learning in EV batteries Applied Energy, 2024, 355:122237

[12] Explainable Data-Driven Digital Twins for Predicting Battery States in Electric Vehicles IEEE Access, 2024

[13] Validation and Implementation in the Cloud of Cell-Level Models of a Digital Twin Simulation Platform IEEE Transactions on Vehicular Technology, 2024